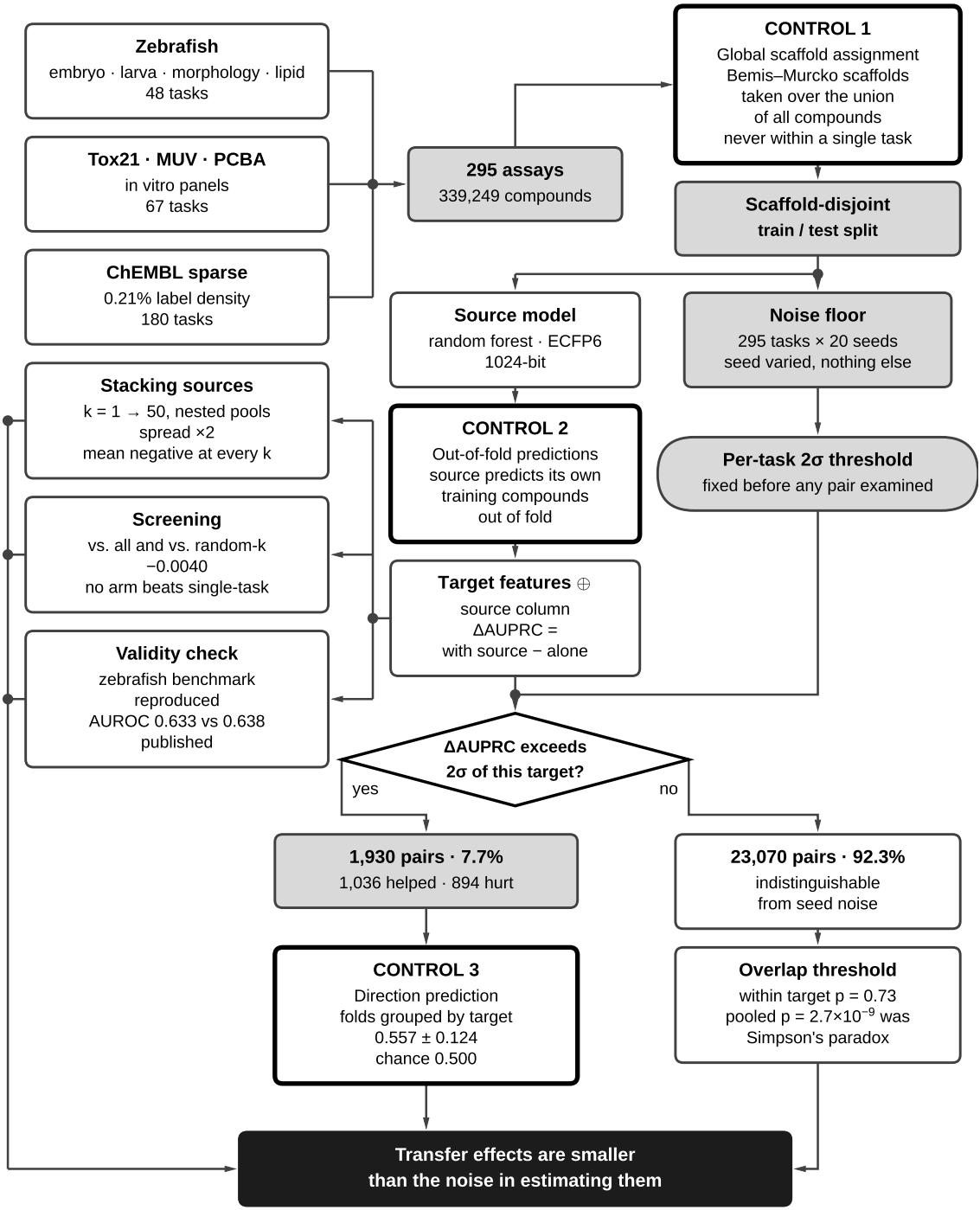
Zebrafish
embryo · larva · morphology · lipid
48 tasks
Tox21 · MUV · PCBA
in vitro panels
67 tasks
ChEMBL sparse
0.21% label density
180 tasks
295 assays
339,249 compounds
CONTROL 1
Global scaffold assignment
Bemis–Murcko scaffolds
taken over the union
of all compounds
never within a single task
Scaffold-disjoint
train / test split
Source model
random forest · ECFP6
1024-bit
Noise floor
295 tasks × 20 seeds
seed varied, nothing else
Stacking sources
k = 1 → 50, nested pools
spread ×2
mean negative at every k
CONTROL 2
Out-of-fold predictions
source predicts its own
training compounds
out of fold
Per-task 2σ threshold
fixed before any pair examined
Screening
vs. all and vs. random-k
−0.0040
no arm beats single-task
Target features ⊕
source column
ΔAUPRC =
with source − alone
Validity check
zebrafish benchmark
reproduced
AUROC 0.633 vs 0.638
published
ΔAUPRC exceeds
2σ of this target?
yes
no
1,930 pairs · 7.7%
1,036 helped · 894 hurt
23,070 pairs · 92.3%
indistinguishable
from seed noise
CONTROL 3
Direction prediction
folds grouped by target
0.557 ± 0.124
chance 0.500
Overlap threshold
within target p = 0.73
pooled p = 2.7×10−9 was
Simpson's paradox
Transfer effects are smaller
than the noise in estimating them